Настройка EC2

Настройка переменных окружения:

```
PS C:\Users\maxpo\yandex-cloud> $env:YC_FOLDER_ID = (yc config get folder-id)
PS C:\Users\maxpo\yandex-cloud> $env:YC_ZONE = (yc config get compute-default-zone)
PS C:\Users\maxpo\yandex-cloud> echo "folder=$env:YC_FOLDER_ID zone=$env:YC_ZONE"
folder=b1g6gpp7fmeo0hlca4ue zone=ru-central1-a
```

Создание сети и подсети:

```
PS C:\Users\maxpo\yandex-cloud> yc vpc subnet create --name devops-lab-subnet-a --network-name devops-lab-net --zone ru-central1-a --range 10.128.0.0/24
id: e9b8boenbptuufcvn7io
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-13T17:55:40Z"
name: devops-lab-subnet-a
network_id: enpj0utadv1hldcqk0pv
zone_id: ru-central1-a
v4_cidr_blocks:
  - 10.128.0.0/24
```

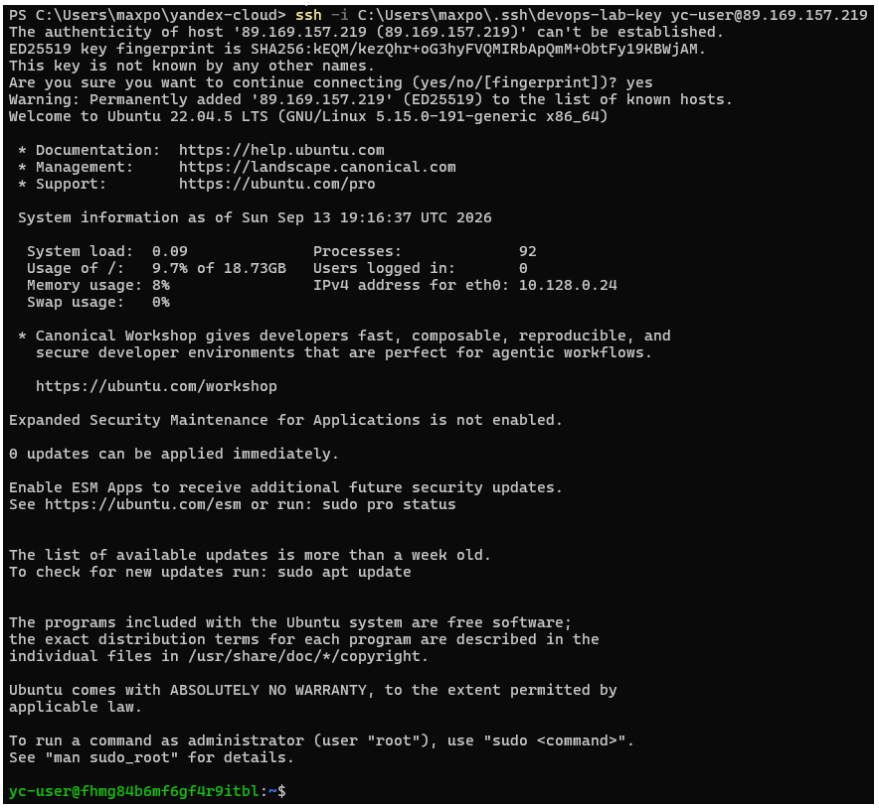
Настройка Security Group:

```
PS C:\Users\maxpo\yandex-cloud> yc vpc security-group update-rules devops-lab-sg --add-rule "direction=ingress,port=80,protocol=tcp,v4-cidrs=0.0.0.0/0"
id: enp552kqevoq0un1tv2a
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-13T18:02:40Z"
name: devops-lab-sg
description: DevOps lab SG
network_id: enpj0utadv1hldcqk0pv
status: ACTIVE
rules:
  - id: enpbc3j87c622068mbj6
    direction: INGRESS
    ports:
      from_port: "22"
      to_port: "22"
    protocol_name: TCP
    protocol_number: "6"
    cidr_blocks:
      v4_cidr_blocks:
        - 37.214.29.120/32
  - id: enp42av13dbirm3jgaqs
    direction: INGRESS
    ports:
      from_port: "80"
      to_port: "80"
    protocol_name: TCP
    protocol_number: "6"
    cidr_blocks:
      v4_cidr_blocks:
        - 0.0.0.0/0
```

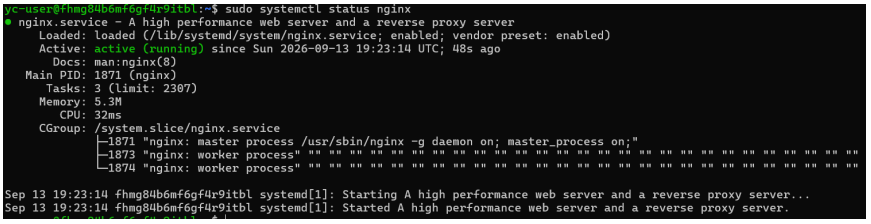
Создание EC2:

```
PS C:\Users\maxpo\yandex-cloud> yc compute instance create --name devops-lab-vm --zone ru-central1-a --network-interface subnet-name=devops-lab-subnet-a,nat-ip-version=ipv4,security-group-ids=enp552kqevoq0un1
tv2a --create-boot-disk image-id=fd8g39rqnaa88u265oic,size=20GB,type=network-ssd --cores 2 --memory 2 --metadata-from-file user-data=./files/user-data.yaml
done (44s)
id: fhmg84b6mf6gf4r9itbl
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-13T19:08:33Z"
name: devops-lab-vm
zone_id: ru-central1-a
platform_id: standard-v2
resources:
  memory: "2147483648"
  cores: "2"
  core_fraction: "100"
status: RUNNING
boot_disk:
  mode: READ_WRITE
  device_name: fhmbhhanfgev10l2v620
  auto_delete: true
  disk_id: fhmbhhanfgev10l2v620
network_interfaces:
  - index: "0"
    mac_address: d0:0d:10:41:16:6b
    subnet_id: e9b8boenbptuufcvn7io
    primary_v4_address:
      address: 10.128.0.24
      one_to_one_nat:
        address: 89.169.157.219
        ip_version: IPV4
    security_group_ids:
      - enp552kqevoq0un1tv2a
fqdn: fhmg84b6mf6gf4r9itbl.auto.internal
scheduling_policy: {}
network_settings:
  type: STANDARD
placement_policy: {}
gpu_settings: {}
metadata_options:
  gce_http_endpoint: ENABLED
  aws_v1_http_endpoint: ENABLED
  gce_http_token: ENABLED
  aws_v1_http_token: DISABLED
  aws_v2_http_endpoint: ENABLED
  aws_v2_http_token: ENABLED
serial_port_settings:
  ssh_authorization: OS_LOGIN
hardware_generation:
  legacy_features:
    pci_topology: PCI_TOPOLOGY_V2
application: {}
```

Подключение к виртуальной машине:

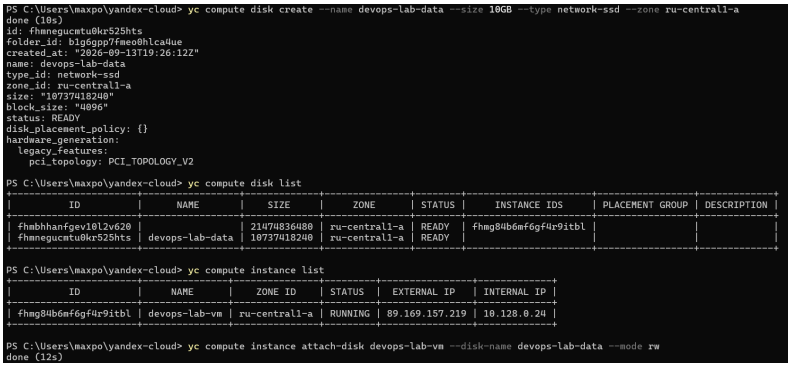


Nginx работает (перед этим добавил ещё одно правило для разрешения исходящего трафика с ec2):



Настройка EBS

Создание и подключение дополнительного диска к VM:



Монтрирование диска на VM:

```
yc-user@fhmg84b6mf6gf4r9itbl:~$ lsblk
NAME    MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
vda     252:0    0   20G  0 disk
├─vda1  252:1    0 19.4G  0 part /
├─vda14 252:14   0    4M  0 part
└─vda15 252:15   0  600M  0 part /boot/efi
vdb     252:16   0   10G  0 disk
yc-user@fhmg84b6mf6gf4r9itbl:~$ sudo mkfs -t ext4 /dev/vdb
mke2fs 1.46.5 (30-Dec-2021)
Discarding device blocks: done
Creating filesystem with 2621440 4k blocks and 655360 inodes
Filesystem UUID: cd5f8161-de4c-43fa-845f-98801c9b3d4c
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376, 294912, 819200, 884736, 1605632

Allocating group tables: done
Writing inode tables: done
Creating journal (16384 blocks): done
Writing superblocks and filesystem accounting information: done

yc-user@fhmg84b6mf6gf4r9itbl:~$ sudo mkdir -p /data
yc-user@fhmg84b6mf6gf4r9itbl:~$ sudo mount /dev/vdb /data
yc-user@fhmg84b6mf6gf4r9itbl:~$ df -h /data
Filesystem      Size  Used Avail Use% Mounted on
/dev/vdb        9.8G   24K  9.3G   1% /data
```

После изменения размера диска:

```
yc-user@fhmg84b6mf6gf4r9itbl:~$ df -h /data
Filesystem      Size  Used Avail Use% Mounted on
/dev/vdb        9.8G   24K  9.3G   1% /data
yc-user@fhmg84b6mf6gf4r9itbl:~$ sudo resize2fs /dev/vdb
resize2fs 1.46.5 (30-Dec-2021)
Filesystem at /dev/vdb is mounted on /data; on-line resizing required
old_desc_blocks = 2, new_desc_blocks = 3
The filesystem on /dev/vdb is now 5242880 (4k) blocks long.

yc-user@fhmg84b6mf6gf4r9itbl:~$ df -h /data
Filesystem      Size  Used Avail Use% Mounted on
/dev/vdb         20G   24K   19G   1% /data
```

Настройка снэпшота

Каждый день в 3:00 UTC, хранить не больше трёх копий:

```
PS C:\Users\maxpo\yandex-cloud> yc compute snapshot-schedule create devops-lab-snap-schedule --expression "0 3 ? * *"
done (4s)
id: fd8tbrlif25g0tvbifbj
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-19T15:52:32Z"
name: devops-lab-snap-schedule
status: ACTIVE
schedule_policy:
  start_at: "1970-01-01T00:00:00Z"
  expression: 0 3 ? * *
snapshot_spec: {}

PS C:\Users\maxpo\yandex-cloud> yc compute snapshot-schedule update devops-lab-snap-schedule --snapshot-count 3
done (2s)
id: fd8tbrlif25g0tvbifbj
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-19T15:52:32Z"
name: devops-lab-snap-schedule
status: ACTIVE
schedule_policy:
  start_at: "1970-01-01T00:00:00Z"
  expression: 0 3 ? * *
snapshot_count: "3"
snapshot_spec: {}

PS C:\Users\maxpo\yandex-cloud> yc compute disk list
+----------------------+-----------------+-------------+---------------+--------+----------------------+-----------------+-------------+
|          ID          |      NAME       |    SIZE     |     ZONE      | STATUS |     INSTANCE IDS     | PLACEMENT GROUP | DESCRIPTION |
+----------------------+-----------------+-------------+---------------+--------+----------------------+-----------------+-------------+
| fhmbhhanfgev10l2v620 |                 | 21474836480 | ru-central1-a | READY  | fhmg84b6mf6gf4r9itbl |                 |             |
| fhmnegucmtu0kr525hts | devops-lab-data | 21474836480 | ru-central1-a | READY  | fhmg84b6mf6gf4r9itbl |                 |             |
+----------------------+-----------------+-------------+---------------+--------+----------------------+-----------------+-------------+

PS C:\Users\maxpo\yandex-cloud> yc compute snapshot-schedule add-disks devops-lab-snap-schedule --disk-id fhmnegucmtu0kr525hts
done (5s)
id: fd8tbrlif25g0tvbifbj
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-19T15:52:32Z"
name: devops-lab-snap-schedule
status: ACTIVE
schedule_policy:
  start_at: "1970-01-01T00:00:00Z"
  expression: 0 3 ? * *
snapshot_count: "3"
snapshot_spec: {}
```

Настройка AMI

Создание VM из образа созданного из диска:

```
PS C:\Users\maxpo\yandex-cloud> yc compute image create --name devops-lab-boot-image-v1 --source-disk-id fhmbhhanfgev10l2v620 --description "Bootable image from devops-lab-vm"
done (34s)
id: fd82m0mlre1jv4plslp
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-19T16:26:11Z"
name: devops-lab-boot-image-v1
description: Bootable image from devops-lab-vm
size: "21474836480"
storage_size: "2977955840"
min_disk_size: "21474836480"
product_ids:
  - f2eevtd24b3fpr7h1ihc
status: READY
os:
  type: LINUX
hardware_generation:
  legacy_features:
    pci_topology: PCI_TOPOLOGY_V2

PS C:\Users\maxpo\yandex-cloud> yc compute instance create --name devops-lab-vm-from-image --zone ru-central1-a --network-interface subnet-name=devops-lab-subnet-a,nat-ip-version=ipv4,security-group-ids=enp55
2kqevoq0un1tv2a --create-boot-disk image-id=fd82m0mlre1jv4plslp,size=20GB,type=network-ssd --cores 2 --memory 2 --ssh-key C:\Users\maxpo\.ssh\devops-lab-key.pub
done (1m0s)
id: fhmi6ic6lla284v4q8s1
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-19T16:27:33Z"
name: devops-lab-vm-from-image
zone_id: ru-central1-a
platform_id: standard-v2
```

Подключение:

```
PS C:\Users\maxpo\yandex-cloud> ssh -i C:\Users\maxpo\.ssh\devops-lab-key yc-user@84.201.134.181
The authenticity of host '84.201.134.181 (84.201.134.181)' can't be established.
ED25519 key fingerprint is SHA256:huCb1GVaMSzqkRmhD3pVMTHBjjfGFAbUN+rQuU2v7TQ.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '84.201.134.181' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-191-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Sep 19 16:29:17 UTC 2026

  System load:  0.23              Processes:             100
  Usage of /:   9.8% of 18.73GB   Users logged in:       0
  Memory usage: 9%                IPv4 address for eth0: 10.128.0.13
  Swap usage:   0%

 * Canonical Workshop gives developers fast, composable, reproducible, and
   secure developer environments that are perfect for agentic workflows.

   https://ubuntu.com/workshop

Expanded Security Maintenance for Applications is not enabled.

4 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


Last login: Sat Sep 19 16:19:18 2026 from 37.214.37.101
yc-user@fhmi6ic6lla284v4q8s1:~$
```

Создание и подключение EFS

```
PS C:\Users\maxpo\yandex-cloud> yc compute filesystem create --name devops-lab-fs --type network-ssd --zone ru-central1-a --size 10GB
done (6s)
id: fhm5iee2v8u80u3dff1t
folder_id: b1g6gpp7fmeo0hlca4ue
created_at: "2026-09-19T16:42:03Z"
name: devops-lab-fs
type_id: network-ssd
zone_id: ru-central1-a
size: "10737418240"
status: READY

PS C:\Users\maxpo\yandex-cloud> yc compute instance stop devops-lab-vm
done (22s)
PS C:\Users\maxpo\yandex-cloud> yc compute instance attach-filesystem --id fhmg84b6mf6gf4r9itbl --filesystem-id fhm5iee2v8u80u3dff1t --device-name shared-fs

done (7s)
```

Монтирование внутри машины:

```
yc-user@fhmg84b6mf6gf4r9itbl:~$ sudo mkdir -p /mnt/shared
yc-user@fhmg84b6mf6gf4r9itbl:~$ sudo mount -t virtiofs shared-fs /mnt/shared
yc-user@fhmg84b6mf6gf4r9itbl:~$ df -T
Filesystem     Type     1K-blocks    Used Available Use% Mounted on
tmpfs          tmpfs       201072     760    200312   1% /run
/dev/vda1      ext4      19644896 1916772  17711740  10% /
tmpfs          tmpfs      1005352       0   1005352   0% /dev/shm
tmpfs          tmpfs         5120       0      5120   0% /run/lock
/dev/vdb       ext4      20531836      24  19550748   1% /data
/dev/vda15     vfat        613184    6232    606952   2% /boot/efi
tmpfs          tmpfs       201068       0    201068   0% /run/user/1000
shared-fs      virtiofs  10485760       0  10485760   0% /mnt/shared
```

Создание и настройка S3

```
PS C:\Users\maxpo\yandex-cloud> yc storage bucket create --name "devops-lab-s3" --default-storage-class standard
name: devops-lab-s3
folder_id: b1g6gpp7fmeo0hlca4ue
anonymous_access_flags: {}
default_storage_class: STANDARD
versioning: VERSIONING_DISABLED
created_at: "2026-09-19T17:20:53.995007Z"
resource_id: e3en77o3jfaba543jefg
```

Проверка, что объект может добавляться и что его можно увидеть:

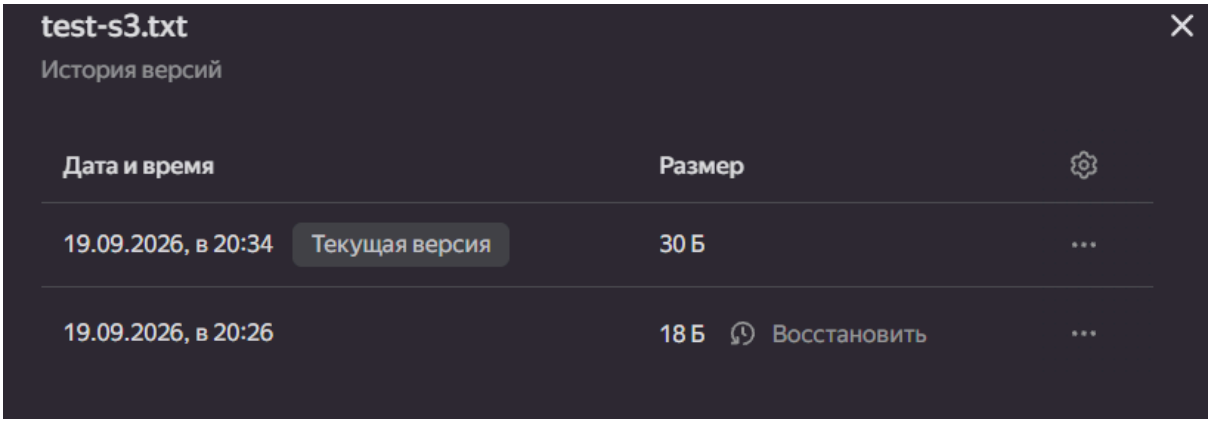
```
PS C:\Users\maxpo\devops-tms\homework35-36> yc storage s3api put-object --body test-s3.txt --bucket "devops-lab-s3" --key prefix/test-s3.txt
etag: '"db70d4aa4086ddab0eb97153f4700fca"'
request_id: 21846cad26df610b
```

```
PS C:\Users\maxpo\devops-tms\homework35-36> yc storage s3api list-objects --bucket "devops-lab-s3" --prefix prefix/
contents:
  - key: prefix/test-s3.txt
    last_modified: "2026-09-19T17:26:06.293Z"
    etag: '"db70d4aa4086ddab0eb97153f4700fca"'
    size: "18"
    owner:
      id: ajen9q47qajgit99hd8t
      display_name: ajen9q47qajgit99hd8t
    storage_class: STANDARD
name: devops-lab-s3
prefix: prefix/
max_keys: "1000"
key_count: "1"
request_id: b5383b301ccdcd46
```

Версионирование:

```
PS C:\Users\maxpo\devops-tms\homework35-36> yc storage s3api put-object --body test-s3.txt --bucket devops-lab-s3 --key prefix/test-s3.txt
etag: '"9bb863e1f36b1e2cc257ed2768105ae5"'
request_id: 99e6b9e456db110c
version_id: 00065BD96F7FF83A
```

```
PS C:\Users\maxpo\devops-tms\homework35-36> yc storage bucket update --name "devops-lab-s3" --versioning versioning-enabled
name: devops-lab-s3
folder_id: b1g6gpp7fmeo0hlca4ue
default_storage_class: STANDARD
versioning: VERSIONING_ENABLED
acl: {}
created_at: "2026-09-19T17:20:53.995007Z"
resource_id: e3en77o3jfaba543jefg
```



Настройка lifecycle:

Welcome | test-s3.txt | lifecycle.json

{} lifecycle.json > [ ] lifecycleRules > { } 0 > { } filter > abc prefix

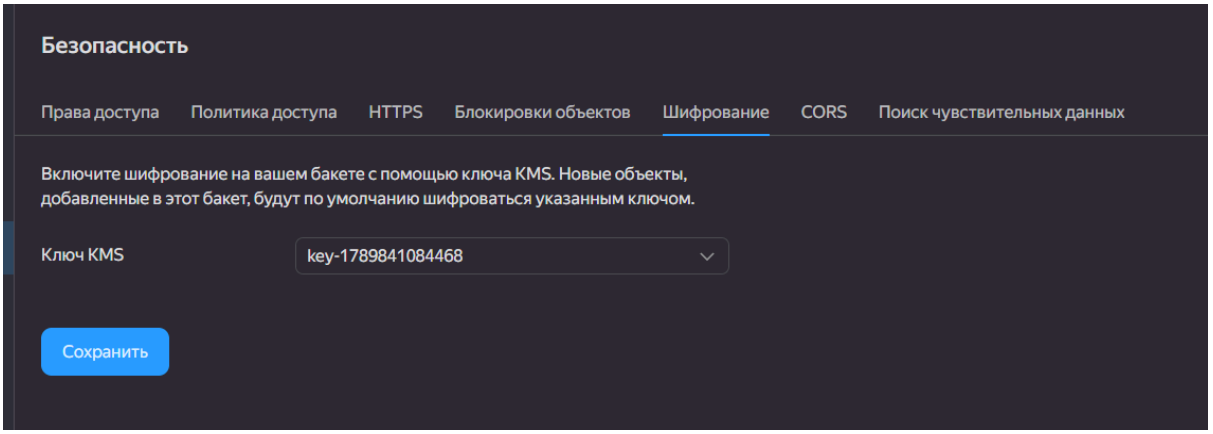
```
{
  "lifecycleRules": [
    {
      "id": "expire-prefix",
      "enabled": true,
      "filter": {
        "prefix": "prefix/"
      },
      "expiration": {
        "days": 7
      }
    }
  ]
}
```

```
PS C:\Users\maxpo\devops-tms\homework35-36> yc storage bucket update devops-lab-s3 --lifecycle-rules-from-file lifecycle.json
name: devops-lab-s3
folder_id: b1g6gpp7fmeo0hlca4ue
default_storage_class: STANDARD
versioning: VERSIONING_ENABLED
acl: {}
created_at: "2026-09-19T17:20:53.995007Z"
lifecycle_rules:
  - id: expire-prefix
    enabled: true
    filter:
      prefix: prefix/
    expiration:
      date: "0001-01-01T00:00:00Z"
      days: "7"
      expired_object_delete_marker: false
resource_id: e3en77o3jfaba543jefg
```

Presigned url:

```
PS C:\Users\maxpo\devops-tms\homework35-36> yc storage s3 presign s3://devops-lab-s3/prefix/test-s3.txt --expires-in 3600 --local --access-key YCAJEF0TKFQt6
oGe-4JP7Cjpq --secret-key YCM_2vEf__t_TyFANbB7agvZjf79iWrXw26rXSYk
https://storage.yandexcloud.net:443/devops-lab-s3/prefix/test-s3.txt?X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=YCAJEF0TKFQt6oGe-4JP7Cjpq%2F20260919%
2Fdefault%2Fs3%2Faws4_request&X-Amz-Date=20260919T180049Z&X-Amz-Expires=3600&X-Amz-SignedHeaders=host&x-id=GetObject&X-Amz-Signature=881ade5b4847698866ec006
f4069df603cf824f5769152597ff9d047d0e14dc0
PS C:\Users\maxpo\devops-tms\homework35-36> curl.exe "https://storage.yandexcloud.net:443/devops-lab-s3/prefix/test-s3.txt?X-Amz-Algorithm=AWS4-HMAC-SHA256&
X-Amz-Credential=YCAJEF0TKFQt6oGe-4JP7Cjpq%2F20260919%2Fdefault%2Fs3%2Faws4_request&X-Amz-Date=20260919T180049Z&X-Amz-Expires=3600&X-Amz-SignedHeaders=host&
x-id=GetObject&X-Amz-Signature=881ade5b4847698866ec006f4069df603cf824f5769152597ff9d047d0e14dc0"
some text for test version 2
PS C:\Users\maxpo\devops-tms\homework35-36>
```

Включение шифрования:



```
PS C:\Users\maxpo\devops-tms\homework35-36> yc kms symmetric-key add-access-binding --role kms.keys.encrypterDecrypter --id abjmoge222q9b05virb5 --service-a
ccount-name devops-lab-admin
done (4s)
```